

Admissible Frontier Saturation and the Cascade Exponent: A Representation-Theoretic Obstruction and its Matrix-Level Refinement

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Abstract

The Cosmochrony relaxation cascade generates an inter-generational mass hierarchy through the contraction of the Kesten–McKay spectral support as the effective valence $p(n)$ grows. The cascade exponent β governing $p(n) \sim n^\beta$ was identified in companion papers as the sole remaining structural parameter of the admissibility sub-programme: O3 constrains $\beta^* \in (0.09, 0.13)$ from lepton masses, while O4 derives the quadratic upper bound $p(n) \lesssim Cn^2$ from bounded-flux and Cheeger arguments, leaving a factor of 10–20 unexplained.

The present paper addresses this gap by introducing the notion of *admissible frontier*: the subset of the raw BFS boundary whose elements introduce genuinely new directions in the admissible spectral subspace. We prove that the vertex-based admissible frontier saturates completely at scale $O(|\text{Cl}(G)|) = O(q)$, far below $|G| = O(q^3)$, through a direct consequence of the representation theory of $G = \text{PSL}(2, \mathbb{F}_q)$. This Proposition is a theorem, independent of any phenomenological input.

We then examine transition-based refinements. Character-based transition fingerprints collapse to the same saturation behaviour on LPS graphs, where all generators lie in a single conjugacy class. Fixed-dimensional matrix proxies produce a clear pre-saturation decay of the admissible frontier fraction but with a q -independent saturation threshold, disqualifying them as structural mechanisms. A Steinberg-based fingerprint using the permutation action on $\mathbb{P}^1(\mathbb{F}_q)$ achieves the first q -structural saturation ($|S^*| \sim q^{1.15}$, $|S^*|/|G| \rightarrow 0$), but the pre-saturation window remains too short to extract a reliable effective exponent.

We conclude that the smallness of β^* is not a representation-theoretic dimension effect but a genuinely dynamical redundancy phenomenon, and identify $\dim \mathcal{H}_{\text{mat}} = O(q^2)$ as the minimal ambient dimension for a structurally complete theory. The correct class of mechanisms is isolated; quantitative extraction of the exponent is deferred to the next paper.

Keywords: admissible frontier; cascade exponent; $\text{PSL}(2, \mathbb{F}_q)$; LPS Ramanujan graphs; Kesten–McKay measure; representation theory; Steinberg representation; mass hierarchy; spectral admissibility.

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1 Introduction

1.1 Background and motivation

The Cosmochrony framework [1] posits that physical observables arise through a non-injective projection $\Pi: \Omega_\chi \rightarrow \mathcal{O}$, where Ω_χ is a pre-geometric relational substrate. The non-injectivity is not a defect but a structural feature responsible for bounded projective capacity, emergent gauge structure, and finite resolution [2].

Effective particle masses emerge from the relaxation dynamics on Ω_χ , organised into a discrete cascade indexed by rank n . At each rank, the connectivity of the substrate is encoded in a Ramanujan–LPS Cayley graph $G(n) = X^{p(n),q}$ of degree $d(n) = p(n) + 1$, where $p(n)$ is the effective valence parameter [7]. As $p(n)$ grows, the Kesten–McKay spectral support contracts toward $\lambda = 1$, causing fixed ADE eigenvalue levels to exit the support at distinct cascade ranks and generating the observed mass ordering [8].

The inter-generational mass ratios are governed by

$$\frac{M_i}{M_j} \approx \left(\frac{|\lambda_i - 1|}{|\lambda_j - 1|} \right)^{2/\beta}, \quad (1)$$

where λ_i are fixed ADE eigenvalues and β is the exponent in $p(n) \sim n^\beta$ [8]. Matching (1) to the charged lepton hierarchy yields $\beta^* \in (0.09, 0.13)$, but leaves β as a phenomenological input.

1.2 What O4 establishes and what remains open

The companion paper O4 [10] shows that the bounded-flux constraint $|\partial_t \chi_v| \leq c_\chi$, inherited from Born–Infeld saturation [2], limits the advance of the exploration front:

$$\Delta p(n) \lesssim c_\chi \sqrt{p(n)}, \quad (2)$$

which integrates to

$$p(n) \lesssim \frac{1}{4} c_\chi^2 n^2. \quad (3)$$

Under the convention $p(n) \sim n^\beta$, this implies $\beta \leq 2$ algebraically. The tighter claim $\beta \leq 1$ of [10] identifies the saturated case as the maximal growth rate and restricts attention to $\beta \in (0, 1]$. In either convention, the factor of 10–20 between this bound and the phenomenological window $\beta^* \approx 0.1$ is left unexplained.

The present paper is O5 in the series. Its objective is to identify which mechanisms can and cannot bridge this gap.

1.3 Main results

The central contributions are the following.

Theorem (Proposition 2.5). The vertex-based admissible span $\Pi_A(S)$ satisfies $\dim \Pi_A(S) \leq \text{rank}(M_{\text{adm}}) \leq |\text{Cl}(G)| = O(q)$ for all $S \subseteq G = \text{PSL}(2, \mathbb{F}_q)$. Beyond a saturation threshold $|S^*| = O(q) \ll |G|$, the vertex-based admissible frontier $\partial^A S$ is empty while the raw frontier ∂S remains $\Omega(\sqrt{|S|})$.

Negative structural results (Section 2.4). Character-based transition fingerprints and fixed-dimensional matrix proxies are insufficient to produce a q -structural saturation of the admissible frontier. The first structurally valid construction is the Steinberg-based fingerprint on $\mathbb{P}^1(\mathbb{F}_q)$, with $|S^*| \sim q^{1.15}$ and $|S^*|/|G| \rightarrow 0$.

Localisation result (Remark 2.16). The smallness of β^* cannot arise from a representation-theoretic dimension effect. It must be a genuinely dynamical redundancy phenomenon in a matrix-level transition space of growing dimension.

1.4 Organisation

Section 2 introduces the admissible frontier framework and proves Proposition 2.5. The transition-based refinements are developed and tested numerically in Sections 2.4–2.5. Section 2.6 derives the structural lesson, and Section 2.7 states the conclusion and opens the next programme.

2 Admissible Frontier Saturation

The central difficulty identified in [10] is the following. The LPS Cayley graph $X^{p,q}$ is Ramanujan: its combinatorial expansion is optimal, and the raw boundary $|\partial S_n|$ grows robustly along any BFS cascade. Yet the argument of [10] establishes only the quadratic upper bound

$$p(n) \lesssim \frac{1}{4} c_\chi^2 n^2, \quad (4)$$

derived by integrating $\Delta p(n) \lesssim c_\chi \sqrt{p(n)}$ under the closure hypothesis $p(n) \propto N(n)$. Under the convention $p(n) \sim n^\beta$ of the O-series [8, 10], this implies $\beta \leq 2$ algebraically. The tighter formulation $\beta \leq 1$ of [10] identifies the saturated case as the maximal admissible growth rate; we adopt this convention here. In either interpretation, the bound leaves the phenomenological window $\beta^* \in (0.09, 0.13)$ of [8] unexplained by a factor of 10–20. The gap cannot be closed by refining the Cheeger bound alone, since the Ramanujan property of $X^{p,q}$ implies that Cheeger constants are already near-optimal for the graph combinatorics.

Remark 2.1 (Decoupling from the O4 normalisation). The quadratic bound $p(n) \lesssim Cn^2$ is the only statement used from [10] in the present work. Any further normalisation of β is purely conventional and does not affect the arguments below.

The resolution proposed here is that the relevant quantity is not the raw boundary ∂S_n but the *admissibly productive* boundary: the subset of ∂S_n whose elements introduce genuinely new directions in the admissible spectral subspace. We show that this quantity saturates rapidly and irrevocably, at a scale $|S^*| = O(|\text{Cl}(G)|) \ll |G|$, independently of the graph’s combinatorial expansion.

2.1 Admissible sectors and the character matrix

Let $G = \text{PSL}(2, \mathbb{F}_q)$ with LPS generating set $\mathcal{S}_p \subset G$ of cardinality $d = p + 1$. We recall from [4] that the bounded-flux constraint $|\partial_t \chi_v| \leq c_\chi$ induces a spectral admissibility envelope

$$A_\rho^{\max} = \frac{c_\chi}{\sqrt{\lambda_\rho}}, \quad (5)$$

where $\lambda_\rho = d - \mu_\rho$ is the Laplacian eigenvalue of the irreducible sector $\rho \in \widehat{G}$. The sector ρ is *admissible* if $0 < \lambda_\rho \leq \lambda^*$, where $\lambda^* = (\sqrt{d-1} + 1)^2$ is the Ramanujan spectral bound [4]. We write $\widehat{G}^{\text{adm}} \subset \widehat{G}$ for the set of admissible sectors.

Definition 2.2 (Admissible character matrix). The *admissible character matrix* is

$$M_{\text{adm}} \in \mathbb{R}^{|\text{Cl}(G)| \times |\widehat{G}^{\text{adm}}|}, \quad (M_{\text{adm}})_{[g], \rho} := \chi_\rho(g),$$

where $[g]$ denotes the conjugacy class of g and χ_ρ is the character of ρ . Rows are indexed by conjugacy classes; columns by admissible sectors.

Remark 2.3 (Dimensions). For $G = \text{PSL}(2, \mathbb{F}_q)$ with q an odd prime: $|G| = q(q^2 - 1)/2 = O(q^3)$; $|\text{Cl}(G)| = O(q)$; $|\widehat{G}^{\text{adm}}| \leq |\widehat{G}| = O(q)$. Hence M_{adm} is $O(q) \times O(q)$ and $\text{rank}(M_{\text{adm}}) \leq |\text{Cl}(G)| = O(q)$, while the group itself has order $O(q^3)$.

2.2 The vertex-based admissible fingerprint

Definition 2.4 (Vertex-based admissible fingerprint). For $g \in G$, the *vertex-based admissible fingerprint* is

$$\pi_A(g) := (\kappa_\rho \chi_\rho(g))_{\rho \in \widehat{G}^{\text{adm}}} \in \mathbb{R}^{|\widehat{G}^{\text{adm}}|},$$

where $\kappa_\rho = \mu_\rho/d$ is the normalised adjacency eigenvalue (*spectral coupling*, [3]). For $S \subseteq G$ define

$$\Pi_A(S) := \text{span}_{\mathbb{R}}\{\pi_A(g) : g \in S\} \subseteq \mathbb{R}^{|\widehat{G}^{\text{adm}}|},$$

and the *vertex-based admissible frontier* $\partial^A S := \{v \in \partial S : \pi_A(v) \notin \Pi_A(S)\}$.

2.3 Rapid saturation of the vertex-based admissible span

Proposition 2.5 (Uniform saturation of the vertex-based admissible span). *Let $G = \text{PSL}(2, \mathbb{F}_q)$, $\widehat{G}^{\text{adm}}$ the admissible sectors, and $\pi_A, \Pi_A(S), \partial^A S$ as in Definition 2.4. Then for any $S \subseteq G$:*

$$\dim \Pi_A(S) \leq \text{rank}(M_{\text{adm}}) \leq |\text{Cl}(G)|. \quad (6)$$

There exists a saturation threshold $S^ \subseteq G$ with $|S^*| \leq \text{rank}(M_{\text{adm}})$ such that once $\dim \Pi_A(S) = \text{rank}(M_{\text{adm}})$, no further element of G can enlarge $\Pi_A(S)$. Consequently:*

$$\partial^A S = \emptyset \quad \text{for all } S \supseteq S^*. \quad (7)$$

Proof. For every $g \in G$, the character $\chi_\rho(g)$ is a class function depending only on $[g] \in \text{Cl}(G)$. Hence $\pi_A(g)$ is constant on each conjugacy class, and the set of all admissible fingerprints is contained in the row space of M_{adm} . This gives $\dim \Pi_A(S) \leq \text{rank}(M_{\text{adm}}) \leq |\text{Cl}(G)|$ for all $S \subseteq G$. Since the row space is finite-dimensional, there exists a finite set S^* with $|S^*| \leq \text{rank}(M_{\text{adm}})$ that already spans the full row space. Once $\Pi_A(S) = \text{rowspan}(M_{\text{adm}})$, every $v \in G$ satisfies $\pi_A(v) \in \Pi_A(S)$, so $\partial^A S = \emptyset$. $\square$

Corollary 2.6 (Admissible productivity bounded by group structure). *Along any BFS cascade $(S_n)_{n \geq 0}$ on $X^{p,q}$:*

$$r_n := \frac{\dim \Pi_A(S_n)}{|S_n|} = O\left(\frac{\text{rank}(M_{\text{adm}})}{|S_n|}\right) \xrightarrow{n \rightarrow \infty} 0. \quad (8)$$

The span $\Pi_A(S_n)$ saturates at $|S_n^| \leq \text{rank}(M_{\text{adm}}) = O(q)$, after which $|\partial^A S_n| = 0$ while $|\partial S_n|$ remains $\Omega(\sqrt{|S_n|})$ by the Cheeger inequality for Ramanujan graphs [10].*

Remark 2.7 (Numerical verification). For $G = \text{PSL}(2, \mathbb{F}_{13})$ with $p = 5, d = 6$: $|G| = 4368$, $|\widehat{G}^{\text{adm}}| = 12$, $\text{rank}(M_{\text{adm}}) = 5$. A BFS cascade on $X^{5,13}$ confirms saturation at $|S^*| \approx 35$, with $|\partial^A S_n| = 0$ for all $|S_n| \geq 38$, while $|\partial S_n| \sim 160\text{--}270$. The ratio r_n decays as $O(5/|S_n|)$. See Figure 1.

Remark 2.8 (Vertex-based bound is a structural upper bound, not the final growth law). The fingerprint $\pi_A(g)$ identifies group elements through conjugacy classes alone, capturing global projective redundancy but not transition-level novelty. Proposition 2.5 is therefore a structural upper bound on admissible productivity, not yet the dynamical growth law for β .

2.4 The transition-based admissible frontier

Proposition 2.5 motivates a finer notion of admissible novelty sensitive to the *transition* by which a boundary vertex is reached.

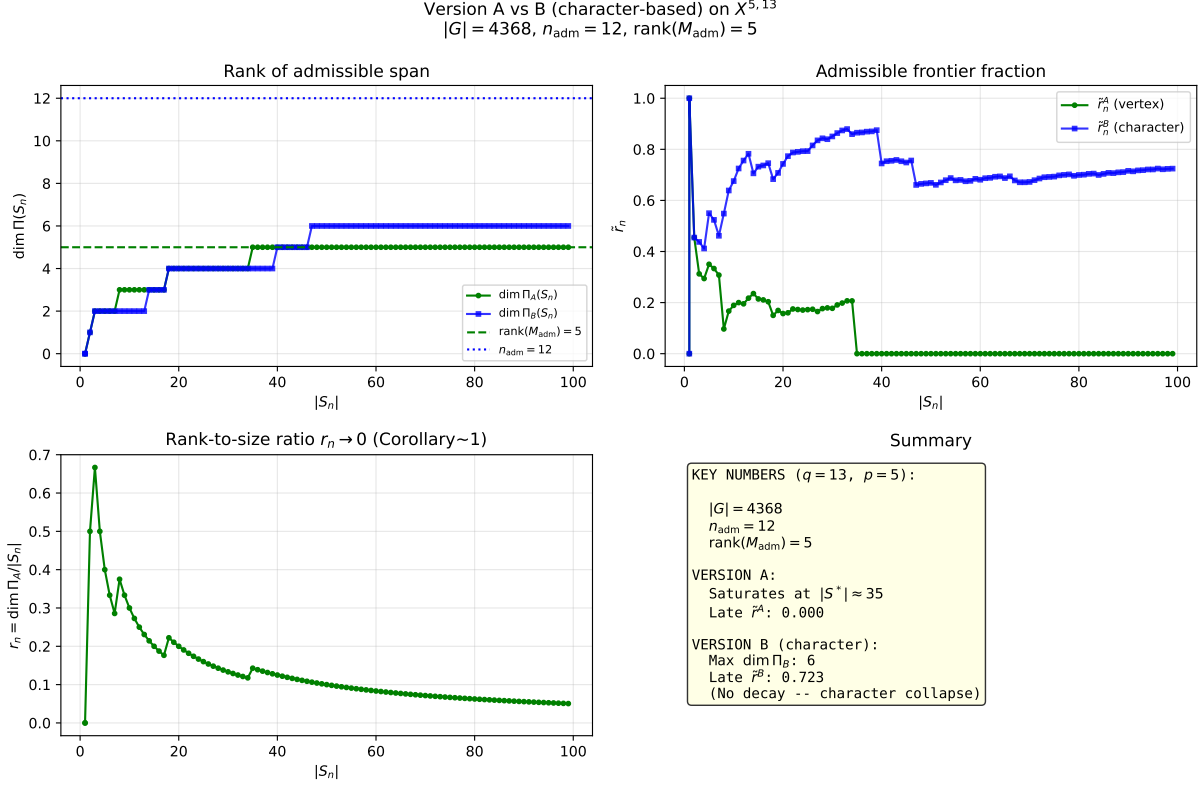


Figure 1: Version A vs. character-based Version B on $X^{5,13}$ ($|G| = 4368$, $n_{\text{adm}} = 12$, $\text{rank}(M_{\text{adm}}) = 5$). *Top left*: $\dim \Pi_A(S_n)$ saturates at rank 5 around $|S_n| \approx 35$ while $\dim \Pi_B(S_n)$ grows to 14 without reaching the ambient cap. *Top right*: $\tilde{r}_n^A$ drops to zero exactly at saturation; $\tilde{r}_n^B$ stays near 0.7 with no visible decay, confirming the character-collapse diagnosis of Remark 2.11. *Bottom*: the rank-to-size ratio $r_n = \dim \Pi_A/|S_n|$ decays as $O(\text{rank}/|S_n|)$, consistent with Corollary 2.6.

Definition 2.9 (Transition-based admissible fingerprint (character version)). For $u \in S_n$ and $v = u \cdot s \notin S_n$ with $s \in \mathcal{S}_p$, define

$$\pi_B(u \rightarrow v) := (\kappa_\rho \chi_\rho(u) \chi_\rho(v))_{\rho \in \widehat{G}^{\text{adm}}} \in \mathbb{R}^{|\widehat{G}^{\text{adm}}|},$$

with $\Pi_B(S_n) := \text{span}_{\mathbb{R}}\{\pi_B(u \rightarrow v) : u \in S_n, v = us \notin S_n, s \in \mathcal{S}_p\}$ and $\partial^B S_n := \{v \in \partial S_n : \exists u, s, \pi_B(u \rightarrow v) \notin \Pi_B(S_n)\}$.

Remark 2.10 (Hierarchy of admissible frontiers). By construction $\partial^A S_n \subseteq \partial^B S_n \subseteq \partial S_n$. Version A is the coarsest (vertices identified by conjugacy class); the raw boundary is the finest.

Remark 2.11 (Limitation of the character-based fingerprint). Definition 2.9 uses only character data. On LPS graphs, the generators $\mathcal{S}_p$ lie in a single conjugacy class, so $\chi_\rho(s)$ is constant across $s \in \mathcal{S}_p$ for every ρ . The fingerprint $\pi_B(u \rightarrow v)$ therefore carries no directional information about the generator s and collapses to essentially the same saturation behaviour as Version A. A relational refinement using representation matrices $\rho(s)$ is required.

Remark 2.12 (Numerical evidence for the character-based limitation). The character-based Version B was tested on $X^{5,13}$ and $X^{5,29}$. The ratio $\tilde{r}_n^B = |\partial^B S_n|/|\partial S_n|$ remains near 0.68 for $X^{5,13}$ and near 0.77 for $X^{5,29}$ after the initial transient, with no visible decay toward zero. This is consistent with the structural diagnosis of Remark 2.11: character collapse on LPS graphs prevents the transition fingerprint from resolving genuine directional novelty. See Figure 2.

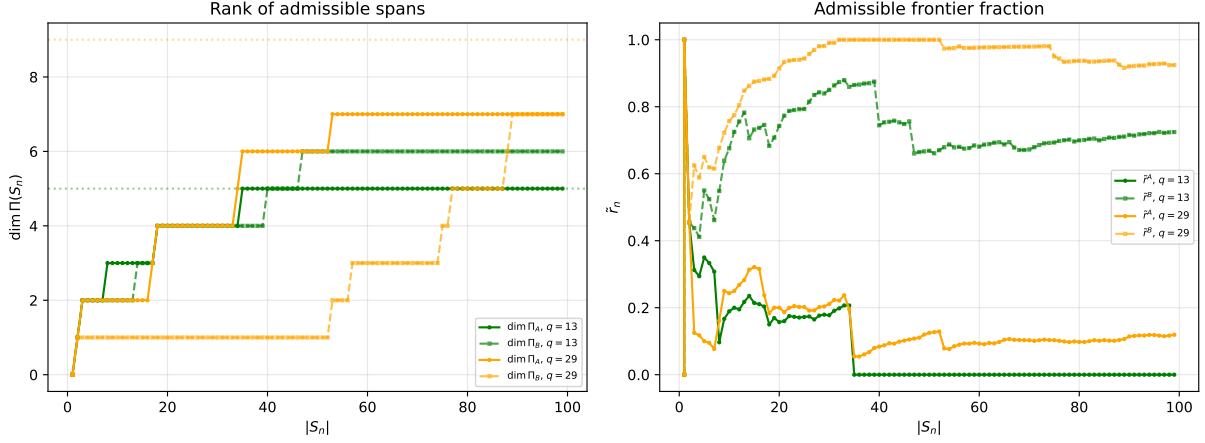


Figure 2: q -dependence of the character-based Version B saturation on $X^{5,q}$ for $q \in \{13, 29\}$. *Left:* $\dim \Pi_A$ saturates rapidly (q -structural), while $\dim \Pi_B$ grows beyond $\text{rank}(M_{\text{adm}})$ but does not reach the ambient cap. *Right:* $\tilde{r}_n^B$ remains near 0.7–0.8 for both values of q , with no q -dependent drift. This confirms that the character-based transition fingerprint produces no q -structural saturation and is therefore insufficient as a mechanism for the cascade exponent.

Remark 2.13 (Matrix-based transition fingerprint and the Steinberg prototype). A structurally adequate transition fingerprint should use representation operators:

$$\Pi_\rho(u, s) \sim \rho(u) \rho(s), \quad v = us.$$

A prototype was implemented via the permutation action of G on $\mathbb{P}^1(\mathbb{F}_q)$ ($q+1$ points). Each element g induces a permutation σ_g ; the Steinberg projection $\tilde{\sigma}_g = \sigma_g - \text{mean}(\sigma_g)$ lies in the q -dimensional Steinberg component. The fingerprint $\pi_B^{\text{St}}(u \rightarrow v) = \tilde{\sigma}_u \odot \tilde{\sigma}_s \in \mathbb{R}^{q+1}$ is $O(q)$ -dimensional and depends on u and s as matrices. Crucially, the LPS generators have six distinct Steinberg projections (while lying in only two conjugacy classes), breaking the character-level degeneracy.

Numerical tests on $X^{5,q}$ for $q \in \{13, 17, 29\}$ yield:

- Saturation threshold $|S_{\text{St}}^*| \approx 16, 21, 40$, scaling as $|S_{\text{St}}^*| \sim q^{1.15}$, growing with q .
- Ratio $|S_{\text{St}}^*|/|G| \sim q^{-1.85} \rightarrow 0$: structurally meaningful.
- Cumulative admissible front: $p_n^{\text{prod}} \sim |S_n|^{1.69}$ (stable, $R^2 > 0.997$), outside $\beta^* \in (0.09, 0.13)$.

The exponent $\beta_{\text{prod}} \approx 1.69$ governs only the pre-saturation regime of this $O(q)$ -dimensional proxy; a representation of larger growing dimension is needed to reach the physically relevant asymptotic regime. See Figures 3 and 4.

Conjecture 2.14 (Matrix-level admissible frontier decay). *There exists a matrix-based definition of the transition-level admissible frontier, refining Definition 2.9, such that*

$$\tilde{r}_n^{B, \text{mat}} := \frac{|\partial^{B, \text{mat}} S_n|}{|\partial S_n|} \xrightarrow{n \rightarrow \infty} 0, \quad (9)$$

and the decay of $\tilde{r}_n^{B, \text{mat}}$ governs the effective cascade exponent via

$$\frac{dp}{dn} \lesssim \tilde{r}_n^{B, \text{mat}} \cdot \sqrt{p(n)}. \quad (10)$$

Remark 2.15 (Scaling implication). Under the power-law Ansatz $\tilde{r}_n^{B, \text{mat}} \sim p(n)^{-\alpha}$, equation (10) integrates to

$$\beta_{\text{eff}} = \frac{1}{1/2 + \alpha}. \quad (11)$$

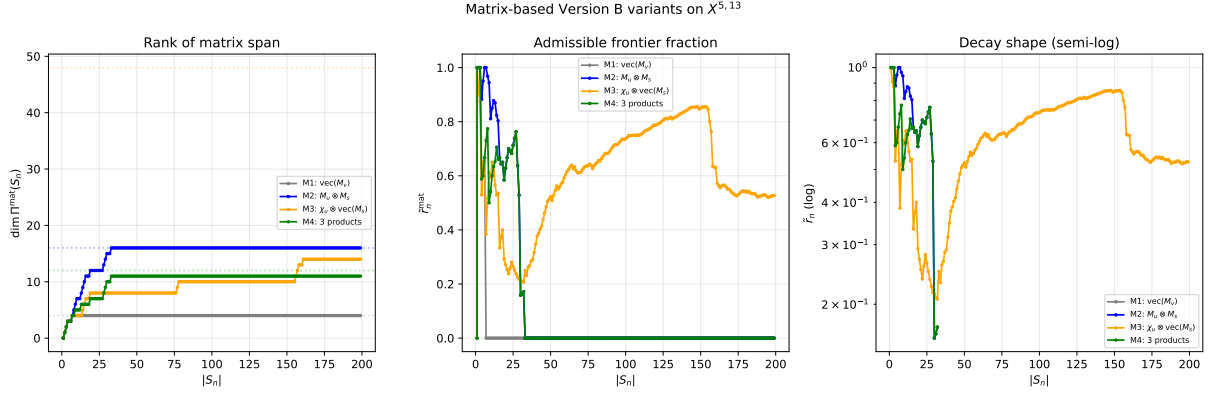


Figure 3: Matrix-based Version B variants on $X^{5,13}$. $M1$ ($\text{vec}(M_v)$, $\dim = 4$): trivial saturation, function of v only. $M2$ ($M_u \otimes M_s$, $\dim = 16$): strong decay to $\tilde{r} = 0$, but q -independent threshold $|S^*| \approx 33$ (fixed ambient dimension artefact). $M3$ ($\chi_\rho(u) \otimes \text{vec}(M_s)$, $\dim = 48$): no decay; character collapse persists as in Version B. $M4$ (three matrix products, $\dim = 12$): strong decay, same artefact as $M2$. Directional matrix structure is *necessary* ($M2/M4$ vs. $M3$) but fixed ambient dimension is *insufficient* ($M2/M4$ vs. Steinberg).

The window $\beta^* \in (0.09, 0.13)$ requires $\alpha \approx 9.5$. A soft polynomial decay is therefore insufficient; the required suppression must be substantially stronger, e.g. a matrix-level saturation-of-support mechanism analogous to Proposition 2.5.

2.5 Summary of contributions

Claim 1 (Structural theorem). Vertex-level admissible novelty saturates at scale $O(q) \ll |G| = O(q^3)$ (Proposition 2.5 and Corollary 2.6). This is proved from the representation theory of $\text{PSL}(2, \mathbb{F}_q)$ alone.

Claim 2 (Negative structural result). Character-based transition fingerprints fail on LPS graphs: the generators lie in a single conjugacy class, so $\chi_\rho(s)$ is constant and no directional information is captured (Remarks 2.11–2.12).

Claim 3 (Negative proxy result). A matrix fingerprint of fixed ambient dimension ($\dim = O(1)$) produces an extinction of $\tilde{r}_n$ but with a q -independent threshold, disqualifying it as a structural mechanism. The Steinberg prototype achieves the first q -structural saturation ($|S^*| \sim q^{1.15}$, $|S^*|/|G| \rightarrow 0$) but at insufficient scale for β_{eff} measurement.

Consequence. The smallness of β^* is not a dimension effect at the character level (Claim 1 is too rigid), nor at the character-transition level (Claim 2), nor at the q -independent matrix level (Claim 3). The correct mechanism is a dynamical redundancy in a matrix-valued space of growing dimension.

2.6 Structural implications

Proposition 2.5 is a group-theoretic fact following from the admissibility filter, with no appeal to dynamics. The dynamical question – why is β small? – is localised precisely in Conjecture 2.14: at what rate does the matrix-level productive frontier decay along the BFS exploration of $X^{p,q}$?

Remark 2.16 (From dimensional saturation to dynamical redundancy). Proposition 2.5 confines vertex-level novelty to a space of dimension $\text{rank}(M_{\text{adm}}) = O(q)$. By contrast, a matrix-level fingerprint $\rho(u)\rho(s)$ lives in $\text{End}(V_\rho)$, so the full matrix-admissible space has dimension

$$\sum_{\rho \in \hat{G}^{\text{adm}}} (\dim \rho)^2 = O(q^3) \sim |G|.$$

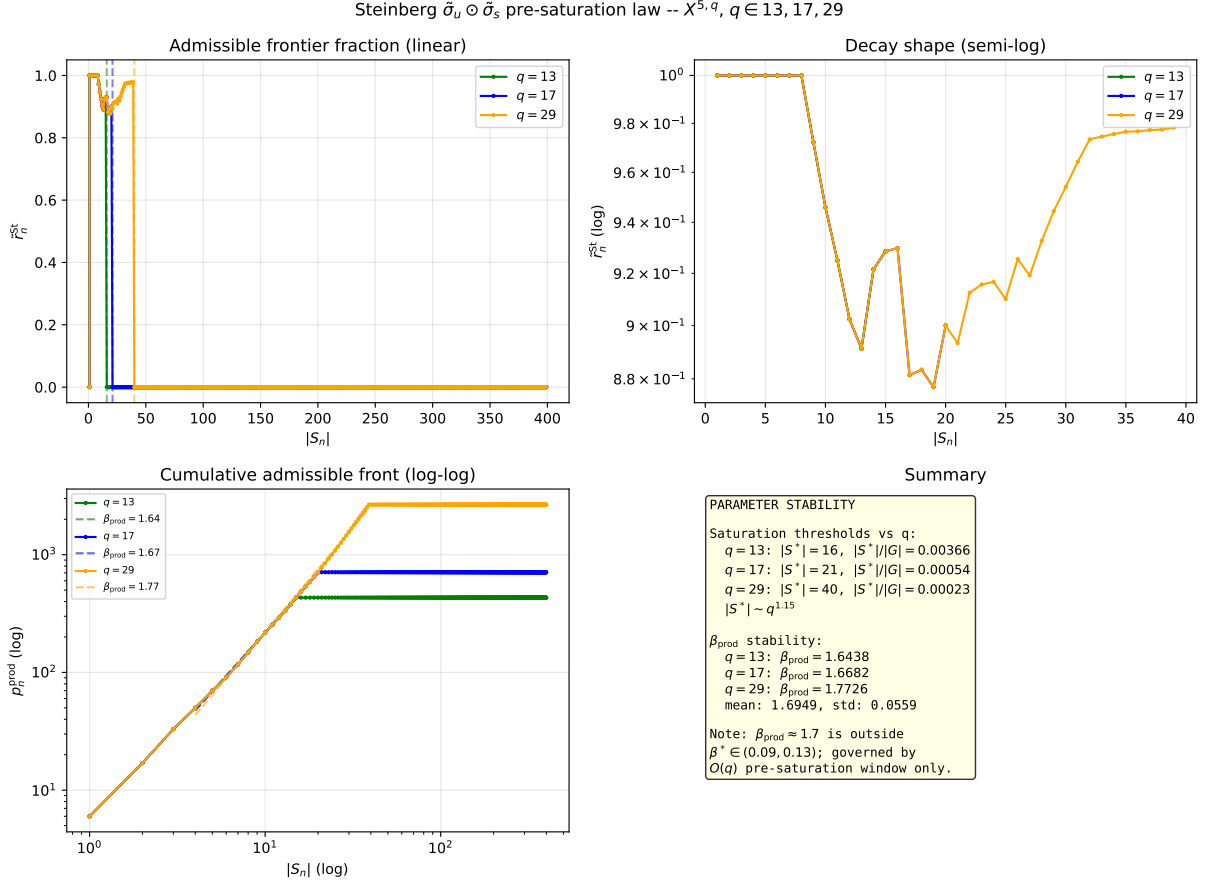


Figure 4: Steinberg-based fingerprint $\pi_B^{\text{St}}(u \rightarrow v) = \tilde{\sigma}_u \odot \tilde{\sigma}_s \in \mathbb{R}^{q+1}$ on $X^{5,q}$ for $q \in \{13, 17, 29\}$ (Remark 2.13). *Top left*: $\tilde{r}_n^{\text{St}}$ decays to zero for all q ; vertical dashed lines mark $|S_{\text{St}}^*|$. *Top right*: semi-log scale shows near-exponential pre-saturation decay. *Bottom left*: cumulative admissible front $p_n^{\text{prod}} \sim |S_n|^{1.69}$ (log-log, stable across q). *Bottom right*: parameter table confirming $|S^*| \sim q^{1.15}$, $|S^*|/|G| \rightarrow 0$, and stable $\beta_{\text{prod}} \approx 1.7$.

No early dimensional saturation is structurally forced for the matrix-based Version B. The suppression of the admissible frontier that would explain β^* *cannot* arise from dimensional constraints at the matrix level; it must originate from a genuinely dynamical mechanism of redundancy. *The smallness of β^* is a dynamical fact, not a representation-theoretic one.*

2.7 Outlook: toward O6

The hierarchy of mechanisms established in this paper is summarised in Table 1.

Conclusion and outlook. The admissible frontier analysis isolates the structural origin of the effective hierarchy left undetermined by O4. At the vertex level, admissible novelty is provably confined to an $O(q)$ -dimensional space, excluding any purely vertex-based explanation of small effective exponents. At the transition level, character-based fingerprints fail on LPS graphs, while fixed-dimensional matrix proxies produce an artificial extinction driven by ambient dimension rather than group structure.

The first structurally consistent construction is the Steinberg-based fingerprint, whose saturation scale grows as $|S^*| \sim q^{1.15}$ with $|S^*|/|G| \rightarrow 0$. However, the pre-saturation regime remains too short to extract a reliable effective exponent, and the observed $\beta_{\text{prod}} \approx 1.7$ lies outside the phenomenological window.

Table 1: Hierarchy of admissible frontier mechanisms

<i>Version</i>	Ambient dim	Status
<i>A (vertex, character)</i>	$O(q)$	Theorem 2.5: saturation proved, window $O(q)$
<i>B (transition, character)</i>	$O(q)$	Insufficient: character collapse on LPS (Remark 2.11)
<i>B (transition, fixed matrix)</i>	$O(1)$	Directional decay, q -independent threshold: not structural
<i>B (Steinberg, $\mathbb{P}^1$)</i>	$O(q)$	First q -structural saturation ($ S^* \sim q^{1.15}$); pre-saturation window too short for β_{eff}

These results collectively demonstrate that the smallness of the cascade exponent cannot be explained by low-dimensional admissible quotients, and must instead arise from a genuinely dynamical mechanism of redundancy in a matrix-level transition space of growing dimension. This identifies the minimal structural requirement for a viable theory of admissible frontier decay, and motivates the next step: the construction of a transition-level fingerprint in a space of dimension $O(q^2)$, where a sufficiently extended pre-saturation regime may allow a stable effective exponent to emerge.

Acknowledgements

The author thanks the Cosmochrony programme for providing the theoretical scaffolding within which these computations are embedded. All numerical experiments were performed in Python using NumPy, SciPy, and Matplotlib.

Data availability

All numerical computations reported here are reproducible from the Python scripts provided as supplementary material. The LPS graph construction follows the explicit generators of [4] (Section 9.6, $p = 5$, $q \in \{13, 17, 29\}$).

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